

Roll No.: _____

Amrita Vishwa Vidyapeetham
Amrita School of Computing, Bangalore
Computer Science and Engineering
Periodical Test 2, November 2022
19CSE353 Mining of Massive Datasets

Duration: 2 hrs

Answer all the questions

Max. Marks: 50

CO	Course Outcome
CO1	Understand the importance of data to business decisions, strategy and behavior. Predictive analytics, data mining and machine learning as tools give new methods for analyzing massive data sets.
CO2	Expose the students to Big data systems like Hadoop, Spark and Hive.
CO3	Explore means to deal with huge document databases and infinite streams of data to mining large social networks and web graphs. Also, learn Algorithms suitable for large scale mining.
CO4	Use of analytic tools, case-studies to provide first hand insight into how big-data problems and their solutions are commercially derived.
CO5	Design Large Scale Machine Learning algorithms with practical hands-on experience for analyzing very large amounts of data.

Answer all questions

- 1.A You have been given a large collections of documents. You want to find “near duplicate” pairs. How does Locality Sensitive Algorithm (LSH) makes it easier to identify observations with various degrees of similarity? Discuss the steps in Locality Sensitive Algorithm in detail. [4 + 4] [BTL 4] [CO3]
- 1.B What is Jaccard Similarity Index? [1 + 1] [BTL 2] [CO3]
 Suppose we have the following two sets of data:
 A = [0, 1, 2, 5, 6, 8, 9]
 B = [0, 2, 3, 4, 5, 7, 9], calculate the Jaccard Similarity.
- 2.A In many data mining situations, we do not know the entire data set in advance. How is Stream Management important when the input rate is controlled externally? Discuss a Twitter Stream Model. [3 + 3] [BTL 3] [CO2]
- 2.B In Twitter media, input elements keep entering at a rapid rate, at one or more input ports (i.e., streams). How do you make critical calculations about the stream using a limited amount of (secondary) memory? How does the system handle this situation? [2 + 2] [BTL 2] [CO2]
- 3.A Discuss the general stream processing model with a neat diagram. [2 + 2] [BTL 2] [CO3]
- 3.B How does Google determine a page's relevance? Discuss the Google Page Rank Algorithm in detail. [2 + 4] [BTL 3] [CO 2]

4.A How do you analyze large graphs? How can a Facebook be viewed as a “Directed Graph”? Explain in detail.

[1 + 3]

[BTL 1]
[CO1]

4.B Consider the below figure 1, calculate the ranks of nodes **y**, **a**, and **m** respectively with detailed steps for each.

[2 +2 +2]

[BTL 2]
[CO2]

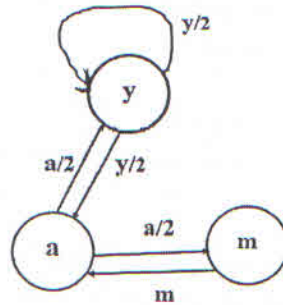


Figure 1

5.A Consider a Transaction database ‘D’ below considering the minimum support count 2 and minimum confidence 60%.

[4 + 4]

[BTL 3]
[CO3]

TID	ITEMS
T1	P1, P2, P3, P5
T2	P1, P2, P3
T3	P1, P2, P4
T4	P1, P2, P5
T5	P2, P4
T6	P2, P3
T7	P1, P3
T8	P2, P3
T9	P1, P3

- Find all frequent itemsets, and
- generate strong association rules from the frequent itemsets using Aprori Algorithm.

5.B Construct a FP-Tree for above database D.

[2]

[BTL 2]
[CO3]

Course Outcome / Bloom’s Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	4	BTL 1	4
CO2	22	BTL 2	18
CO3	24	BTL 3	20
CO4	0	BTL 4	08
CO5	0	BTL 5	0